# Regular Season Scheduling Optimization for the National Basketball Association

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## 1 Problem Statement

### 1.1 Overview

Basketball has emerged as one of the most widely followed sports globally over the past few decades, yet its developmental trajectory indicates substantial remaining growth potential. Originating as a predominantly national pastime, it has progressively expanded its presence across multiple continents, including Europe, Asia, and Africa. In this context, the National Basketball Association (NBA) must reconsider and systematically restructure its game scheduling practices in order to better accommodate its increasingly international fan base, further consolidate and expand its global platform, and potentially approach or surpass the popularity levels achieved by other major sports, such as American football.

One factor that may be constraining the growth of basketball's popularity is the health and availability of its players—particularly its elite athletes. In the 1990s, for instance, all-star players missed on average slightly more than 10 games per 82-game regular season. Approximately three decades later, this figure has risen to about 24 missed games per all-star player per 82-game season. Consequently, some of the most prominent players in the sport are now absent for roughly 30% of the regular season.

This dynamic is particularly detrimental in the context of basketball, a sport that is inherently star-centric. Each team only has five players on the court at a time, and there are effectively no constraints on the proportion of offensive possessions that can be initiated or completed by a single athlete. Consequently, high-impact players frequently account for a disproportionate share of team possessions. For example, high-usage offensive stars can attain usage rates of approximately 35%, a metric that quantifies the proportion of a team's offensive possessions that a player individually terminates by attempting a field goal, drawing free throws, or committing a turnover while on the court. When approximately 35% of all such possessions are in some way mediated by an all-star caliber player, that individual exerts a dominant influence on the team's overall performance potential.

These all-stars are missing games because injuries are becoming more prolific. Coaching staffs have opted to give players more intentional rest during games and across the season to ensure they are properly recovered and ready to endure 48 minutes of intense action. With that, the most interesting pieces of teams are often not even playing in the games, which disincentivizes fans from coming out to support their favorite teams because their favorite players are not around. Imagine, for example, saving up enough money to splurge on tickets to a game and then learning that your favorite player is injured and unable to play. While the game may still be interesting, the motivation to show up will diminish and therefore engagement and revenue will decrease as well.

Ultimately, our objective is to promote the continued expansion of the league and to cultivate a community

of individuals with a shared interest in the sport. At the same time, it is important to recognize that the NBA operates as a commercial enterprise that must generate profit. Consequently, we sought to address the problem of maximizing fan engagement and overall popularity—and, by extension, league revenue—while simultaneously safeguarding player health.

Our primary strategy for supporting player well-being was to constrain the travel distance required of teams immediately following back-to-back games (i.e., games scheduled on consecutive days without a rest day in between). We did not aim to eliminate back-to-back games entirely, as certain high-profile matchups scheduled on specific dates may be critical for viewership and revenue. Instead, our premise was that minimizing post-game travel distance on the second game of a back-to-back sequence would reduce time spent in transit (e.g., on airplanes or buses), thereby increasing the time available for sleep and physiological recovery.

### 1.2 Problem Type

This project is fundamentally problem-driven, as we selected a real-world decision-making challenge, translated it into a mathematical framework, and approached it through optimization methods. Our goal was to construct an objective function that balances two priorities: maximizing the popularity score of selected game matchups on a given day and minimizing the total travel burden associated with back-to-back games. To quantify popularity, we developed a composite metric using standardized and weighted components, including the number of all-star players on each team, social media following, national television exposure, franchise championships, and both city and metropolitan population. These values were manually designed and calibrated by our team. The following sections provide a more detailed explanation of how each component was constructed and incorporated into the model.

### 1.3 Why Does It Matter

This problem is important because it directly influences the long-term sustainability and growth trajectory of the NBA. Although the league has already achieved substantial success, its potential for expansion—both domestically and internationally—remains far greater. An improved scheduling framework can support this growth by enhancing fan engagement, increasing revenue opportunities, and ultimately strengthening the financial health of the league. In turn, this helps ensure competitive player compensation and motivates team owners to continue investing in and promoting the NBA's development.

There are many occasions when there are lulls in the NBA season: durations of time in which casual fans feel like the games do not amount to anything in the short term because they occur in the middle of a long 82-game season and do not directly dictate a playoff spot or elimination. During these periods, there is a lack of engagement and therefore a lack of revenue—a massively missed opportunity. We aim to keep engagement with the league consistent and more evenly spread across the course of the long season.

## 2 Data

Our project relied on several key datasets, the primary one being the complete schedule for the 2024–2025 NBA season. We chose this season as the basis for our analysis because it was the most recent full season available at the time of the project, allowing us to observe all matchups, rest patterns, and travel sequences in their entirety.

The scheduling data was sourced directly from the NBA's official website and provided game-level information in a structured format, including the date, home team, and away team for each matchup. For modeling purposes, we indexed the season from Day 1 through Day 174. We also examined days on which

the NBA intentionally scheduled no games—such as Thanksgiving and the All-Star break—to ensure that our optimized schedule respected these real-world constraints. This allowed the model to remain consistent with league operations and avoid generating games on blackout dates.

The teams were indexed from 1 to 30 in alphabetical order, with the Atlanta Hawks labeled as 1 and the Washington Wizards as 30. This indexing choice allowed us to implement cleaner and more consistent binary decision variables across teams and days. We also assigned each team to its respective conference (East or West) and division (Atlantic, Central, Southeast, Northwest, Pacific, Southwest). These classifications were incorporated into the model to ensure that divisional and conference opponents faced each other the league-mandated number of times.

To capture travel requirements, we constructed a $30 \times 30$ distance matrix representing the mileage between every pair of teams' home arenas. All distances were obtained using Google Maps to provide a standardized approximation of actual travel. This matrix played a critical role once constraints were added to cap the distance a team could travel following a back-to-back sequence, ensuring that our optimization respected realistic fatigue and logistics considerations.

The existing schedule was then encoded into a tensor in Julia, and we created a corresponding empty tensor to store the optimized schedule generated by our model. This structure allowed us to represent all possible team matchups across days in a consistent, high-dimensional format suitable for our constraint and objective formulations.

Beyond these components, relatively little additional data preparation was required. The raw scheduling information was already clean and structured, and the supplementary metrics were developed directly by our team. The following sections provide a detailed explanation of how these metrics were constructed and integrated into the optimization framework.

# 3 Method

## 3.1 Metric Development

### 3.1.1 Overview

To assess the quality of the schedules produced by the optimization model, metrics were developed to align directly with the quantities optimized and constrained in the formulation. Although the proposal initially described a broader revenue-oriented framework, the practical evaluation centered on three outcomes: the objective value, the number of back-to-back (B2B) games, and compliance with the 1600-mile travel restriction. These metrics allow us to compare the optimized schedule to the actual 2024–2025 NBA regular-season schedule under a consistent evaluation rule.

### 3.1.2 Objective Value Metric

The primary measure of schedule quality is the model's objective value across all scheduled matchups.

$$\sum_{i=1}^{30} \sum_{j=1}^{30} \sum_{d=1}^{174} w_{ij}^{d} \, x_{ij}^{d}$$

These weights encode factors such as team popularity, day-of-week weights, and special event days, serving as a proxy for the "value" of assigning a particular matchup to a particular date. Because the actual 2024–2025 schedule can be evaluated using the same weight tensor, objective values are directly comparable.

A higher objective value indicates that an optimized schedule places more high-value games on favorable dates while respecting all constraints.

#### 3.1.3 Back-to-Back Game Count

The total number of back-to-back (B2B) games is computed directly from the generated schedule. For each team $t$, every day the team plays, whether that game is at home or away, are extracted and grouped per day. After sorting these days in increasing order from 1-174, back-to-back games are identified via observing when a given team plays both on day $d$ and day $d+1$. Formally, define

$$G_t(d) = \begin{cases} 1, & \text{if team } t \text{ plays exactly one game on day } d, \\ 0, & \text{otherwise.} \end{cases}$$

Then the number of back-to-backs for team $t$ is

$$\mathrm{B2B}(t) = \sum_{d=1}^{173} \mathbf{1}\{G_t(d) = 1 \ \wedge \ G_t(d+1) = 1\},$$

and the league-wide total is

$$\text{Total B2Bs} = \sum_{t=1}^{30} \mathrm{B2B}(t).$$

This method corresponds to the procedure implemented in the evaluation code in Julia where each team's set of game days is identified, any instances of multiple games on the same day are flagged, and all valid consecutive-day pairs are counted. The resulting metric captures the degree of schedule compression present in the final calendar and serves as an indicator of potential player fatigue.

#### 3.1.4 Mileage Constraint for Back-to-Back Games

In addition to counting the total number of back-to-back (B2B) games, each B2B is evaluated under a travel-distance constraint designed to limit excessive travel between consecutive game days. For every team $t$, if a game on day $d$ is followed by a game on day $d+1$, the corresponding arena-to-arena distance is computed. Let $\mathrm{dist}_{ab}$ denote the travel distance between arenas $a$ and $b$. A B2B is considered travel-feasible when

$$\mathrm{dist}_{ab} < 1600,$$

where $a$ and $b$ denote the locations of the games on days $d$ and $d+1$, respectively. Any B2B requiring travel at or above this threshold is rendered infeasible through the model's scheduling constraints.

For evaluation, the number of B2Bs that satisfy the distance restriction is reported, and the schedule is validated to ensure that no B2Bs violate the 1600-mile limit. This metric distinguishes between simply consecutive games and consecutive games that impose substantial travel demands, therefore providing a more refined measure of fatigue-related schedule quality.

#### 3.1.5 Evaluation Framework and Summary

The metrics described above provide a consistent basis for comparing the optimized schedules with the actual 2024–2025 NBA regular-season schedule. A schedule is regarded as an improvement when three criteria are satisfied. First, the achieved objective value must exceed the objective value obtained when the real schedule is evaluated under the same weight tensor $w_{ij}^d$; second, the number of back-to-back occurrences must comparable to (defined as within about 200-250 games of the actual season), the number observed in

the actual schedule; third, all identified B2Bs must comply with the 1600-mile travel restriction, ensuring that no consecutive-day pairing imposes excessive travel demands.

This evaluation framework aligns directly with the structure of the optimization model and the constraints imposed on the scheduling process. By applying these criteria, schedule quality can be assessed in a manner that reflects both the optimization objectives and the operational considerations of league scheduling.

## 3.2 Modeling

### 3.2.1 Decision Variables

To generate a feasible NBA regular-season schedule, we formulate a mixed-integer optimization model in `Julia` using `JuMP` with the `Gurobi` solver. The core decision variable is:

$$x_{ijd} = \begin{cases} 1 & \text{if away team } i \text{ plays at home team } j \text{ on day } d, \\ 0 & \text{otherwise.} \end{cases}$$

Indices are defined as follows:

- $i, j \in \{1, \ldots, 30\}$ — NBA teams (away/home).
- $d \in \{1, \ldots, 174\}$ — days in the regular season.

We ensure that there is no self-games via $x_{iid} = 0$. To support additional structural constraints (e.g., back-to-backs), we define two auxiliary binary variables:

1. **Game-presence variable**
$$g_{td} = 1 \text{ if team } t \text{ plays on day } d.$$
2. **Back-to-back indicator**
$$b2b_{td} = 1 \text{ if team } t \text{ plays on days } d \text{ and } d+1.$$

These variables are linked to the primary $x$-variables through linear constraints to maintain MIP tractability.

### 3.2.2 Objective Function

The primary scheduling objective maximizes our weight tensor $w_{ijd}$, which encodes the "value" of a game. The objective is:

$$\max \sum_{i=1}^{30} \sum_{j=1}^{30} \sum_{d=1}^{174} w_{ijd}\, x_{ijd}.$$

### 3.2.3 Structural Scheduling Constraints

**(a) Daily game limit per team** Each team may play at most one game per day:

$$\sum_{j \neq t} x_{tjd} + \sum_{i \neq t} x_{itd} \leq 1 \quad \forall t, d.$$

**(b) Season length requirement** Each team must play exactly 82 games:

$$\sum_{j \neq t,\, d} x_{tjd} + \sum_{i \neq t,\, d} x_{itd} = 82 \quad \forall t.$$

**(c) Required home/away matrix** The NBA prescribes a number of games between every pair of teams, but it differs every year. For simplicity, we fix the number of games played by every pair of teams using last year's schedule. Let $G_{ij}$ denote the required count of games with $i$ as away and $j$ as home. We enforce:

$$\sum_d x_{ijd} + x_{jid} = G_{ij}, \quad \forall i < j.$$

This ensures divisional and inter-conference balance.

#### 3.2.4 Calendar-Specific Constraints

**(a) No-game days** Designated dates (e.g., Election Day, Thanksgiving Day, All-Star break) must contain no games:

$$\sum_{i \neq j} x_{ijd} = 0 \quad \forall d \in D_{\text{no-game}}.$$

**(b) Christmas and opening night caps**

$$\sum_{i \neq j} x_{ij,\,\text{Christmas}} \leq 5, \qquad \sum_{i \neq j} x_{ij,\,\text{Opening}} \leq 2.$$

#### 3.2.5 Weekly Load Constraints

The league's competitive-balance guidelines require teams to avoid overly dense weeks. For each team $t$ and week $w$:

$$0 \leq \left( \sum_{d \in w} g_{td} \right) \leq 5.$$

Weeks are defined as 25 contiguous 7-day intervals covering the 174-day season.

#### 3.2.6 Daily League-Wide Game Volume

On any regular (non-special) day, the total number of games must fall between 4 and 10:

$$4 \leq \sum_{i \neq j} x_{ijd} \leq 10 \quad \forall d \in D_{\text{normal}}.$$

This prevents oversaturation or undersupply on the national broadcast calendar.

#### 3.2.7 Consecutive-Game Constraints

To avoid excessive fatigue, the NBA prohibits three games on three consecutive days:

$$g_{td} + g_{t,d+1} + g_{t,d+2} \leq 2 \quad \forall t,\ d = 1, \ldots, 172.$$

This captures all rolling 72-hour windows.

#### 3.2.8 Back-to-Back Limits

A maximum of one back-to-back set per team per week is permitted:

$$\sum_{d \in w} b2b_{td} \leq 1, \quad \forall t,\ w.$$

The indicator variable $b2b_{td}$ is enforced via the linearization:

$$\begin{aligned} b2b_{td} &\geq g_{td} + g_{t,d+1} - 1, \\ b2b_{td} &\leq g_{td}, \\ b2b_{td} &\leq g_{t,d+1}. \end{aligned}$$

#### 3.2.9 Travel-Mileage Restrictions

Using a distance matrix $\text{dist}_{ab}$, we incorporate travel-fatigue constraints preventing teams from undertaking extreme trips ($> 1600$ miles) between consecutive days.

**(a) Away → Away**

$$x_{iad} + x_{ib,d+1} \leq 1 \quad \text{if } \text{dist}_{ab} \geq 1600.$$

**(b) Away → Home**

$$x_{iad} + \sum_{k \neq i} x_{ki,d+1} \leq 1 \quad \text{if } \text{dist}_{ai} \geq 1600.$$

**(c) Home → Away**

$$\sum_{k \neq i} x_{kid} + x_{ib,d+1} \leq 1 \quad \text{if } \text{dist}_{ib} \geq 1600.$$

#### 3.2.10 Warm Start

To accelerate convergence, the model is initialized with a historical, league-feasible schedule $T_{ijd}$:

$$x_{ijd}^{(0)} = T_{ijd}.$$

This warm start significantly improves branch-and-bound performance in high-dimensional scheduling spaces and reduces computational time.

# 4 Results

We now evaluate the performance of the three schedules under consideration: (1) the original 2024–2025 NBA schedule, (2) the optimized schedule using our objective function without any distance restriction on back-to-back games, and (3) the optimized schedule with a 1600-mile travel constraint imposed on the second night of a back-to-back. For each schedule, we examine the resulting objective value, the number of back-to-back sequences generated, and the average travel distance associated with those sequences.

## 4.1 Objective Values Across Schedules

| **Schedule** | **Objective Value** |
|---|---|
| Original 2024–2025 Schedule | 562.715 |
| Optimized (No Distance Constraint) | 641.100 |
| Optimized (1600-Mile Constraint) | 638.600 |

Table 1: Objective values achieved by each schedule variant.

Using our popularity-based objective function, the original 2024–2025 schedule achieved a total score of 562.715. It is important to note that this score reflects weights that we constructed independently of the

league's actual scheduling process; the NBA did not design its schedule to optimize for these specific metrics. When we generated a new schedule using our optimizer without enforcing travel limits, the objective value increased to 641.1—an improvement of approximately 13.9% relative to the baseline. Incorporating the 1600-mile cap on back-to-back travel produced a schedule with an objective value of 638.6, representing a 13.5% improvement over the original schedule and only a 0.4% reduction compared to the unconstrained optimal solution. This slight decrease is expected, as imposing a travel limit removes certain highly weighted matchups that would otherwise boost the objective value but require excessive travel, thereby constraining the optimizer in favor of more realistic player-health considerations.

## 4.2 Back-to-Back Frequency and Structural Constraints

| Schedule | Number of Back-to-Backs |
|---|---|
| Original 2024–2025 Schedule | 450 |
| Optimized (No Distance Constraint) | 697 |
| Optimized (1600-Mile Constraint) | 686 |

Table 2: Back-to-back counts generated across schedule variants.

Next, we analyze the number of back-to-back sequences generated in each schedule. In the original 2024–2025 season, there were 450 back-to-back games, averaging roughly 15 per team. When we solved the scheduling problem using our objective function without any travel restrictions, the resulting schedule produced 697 back-to-backs, or about 23 per team. This increase was expected: our popularity metric placed substantial weight on weekend games, under the assumption that fan engagement is typically highest on Fridays, Saturdays, and Sundays. As a result, the optimizer naturally attempted to cluster games on weekends, which in turn increased the frequency of back-to-backs. To prevent this behavior from becoming excessive or unrealistic, we implemented additional structural constraints limiting teams to at most one back-to-back per week and prohibiting any team from playing three consecutive days. We also required that each permissible day in the season include between 4 and 10 games, ensuring an even distribution of league-wide activity.

In the schedule generated under the 1600-mile travel constraint, we observed a total of 686 back-to-backs—only 11 fewer than in the unconstrained solution. Although we initially anticipated a more substantial reduction, the model instead preserved many of the same weekend-based patterns but adjusted the pairings to keep the second game of each back-to-back geographically closer to the first. This indicates that the distance constraint shaped where back-to-backs occurred rather than how many were created, reflecting the optimizer's ability to satisfy popularity-driven objectives while still respecting travel limitations.

## 4.3 Travel Distance Analysis

| Schedule | Total B2B Travel (Miles) | Avg. Distance per B2B (Miles) |
|---|---|---|
| Original 2024–2025 Schedule | 476,268 | 1,058 |
| Optimized (No Distance Constraint) | 768,216 | 1,102 |
| Optimized (1600-Mile Constraint) | 556,595 | 811 |

Table 3: Travel distances associated with back-to-backs across schedules.

Analyzing the total distance traveled after a back-to-back, we note that the original schedule had the least distance traveled at 476,268 miles, compared to 768,216 miles in our unconstrained solution and 556,595 miles in our distance-constrained solution. However, this difference is largely due to the sheer number

of back-to-backs: fewer back-to-backs naturally lead to less total distance. When comparing the *average* distance traveled per back-to-back, our distance-bounded model averaged 811 miles, compared to 1058 miles in the actual NBA schedule. This represents roughly a 24% decrease in travel distance while still achieving a 13% higher objective score. This demonstrates that our model achieved precisely what we intended: higher revenue and popularity with less wear-and-tear-inducing travel.

### 4.4 High-Profile Dates: Opening Night and Christmas Day

| **Showcase Day** | **Teams Selected by Optimizer** |
|---|---|
| Opening Night | BOS, GSW, LAL, NYK |
| Christmas Day | BOS, NYK, BKN, GSW, DAL, LAC, PHX, LAL, CHI, MIL |
| **NBA Teams Not Selected by Model** | |
| Opening Night | MIN |
| Christmas Day | MIN, SAS, PHI, DEN |

Table 4: Comparison of optimizer-selected teams and NBA-selected teams for high-profile dates.

Several interesting insights emerged when examining the most prominent days of the NBA season. Opening Night and Christmas Day traditionally represent the league's highest-profile showcase dates, typically featuring a small number of marquee matchups—two games on Opening Night and five on Christmas Day. In both the unrestricted and distance-restricted optimized schedules, the solver consistently selected the same set of highly popular teams for these slots. For Opening Night, the chosen teams were the Boston Celtics, Golden State Warriors, Los Angeles Lakers, and New York Knicks. For Christmas Day, the selected teams were the Boston Celtics, New York Knicks, Brooklyn Nets, Golden State Warriors, Dallas Mavericks, Los Angeles Clippers, Phoenix Suns, Los Angeles Lakers, Chicago Bulls, and Milwaukee Bucks. These teams largely represent major media markets, long-standing historical success, strong brand recognition, and high-profile star talent, all of which naturally align with the weights used in our popularity metric.

When comparing our results to the NBA's actual selections, we observed only minor deviations. The real Opening Night slate differed by a single team, featuring the Minnesota Timberwolves instead of the Golden State Warriors. On Christmas Day, the NBA included the Minnesota Timberwolves, San Antonio Spurs, Philadelphia 76ers, and Denver Nuggets—teams that did not appear in our optimized version. This divergence can largely be attributed to differences in evaluation criteria. While these teams boast emerging superstar talent and, in some cases, recent championship success, their broader market size, historical legacy, and overall popularity scores are not as strong as those of the large, established franchises prioritized by our algorithm. This does not indicate that the league's selections were inferior; rather, it suggests that the NBA incorporates additional qualitative factors—such as narrative-driven matchups, rising-star visibility, and market diversification—whereas our optimization model strictly followed the popularity metrics we defined.

## 5 Impact and Future Steps

### 5.1 Impact

The optimization results demonstrate that meaningful improvements to the NBA regular-season schedule can be achieved under realistic operational constraints. The generated schedule shows that higher objective values, corresponding to more desirable matchup placements in the calendar, can be attained while simultaneously reducing the travel distance required between arenas for back-to-back games. This indicates that schedule quality, as measured by the model's weighting structure, does not need to be sacrificed in order to achieve reductions in travel-induced fatigue.

By reducing the number of demanding back-to-back sequences and limiting long-distance travel between these games, the resulting schedules have the potential to reduce player fatigue and the likelihood of injury. A reduction in injury incidence, particularly among high-profile players, would enhance competitive balance and improve fan engagement across the league. Increased player availability and more strategically placed high-value games may also contribute to higher levels of viewership, attendance, and overall league popularity.

These findings suggest that data-driven scheduling frameworks can support league-wide objectives related to player health, competitive fairness, and global fan engagement. As professional sports leagues continue to expand their reach, tools of this type can offer a practical path toward more efficient and sustainable scheduling practices.

## 5.2 Future Steps

Several extensions of the current modeling framework would allow for improved accuracy, additional flexibility, and closer alignment with real-world scheduling considerations:

- **Refinement of the weight tensor**: The weights $w_{ij}^d$ could be enhanced using more detailed historical data on television ratings, attendance, team performance trajectories, and star-player availability. Learning these weights from more data would allow the objective function to more closely represent league priorities.
- **Direct control of back-to-back frequency**: The current model incorporates mileage-based restrictions but does not impose explicit upper bounds on the number of back-to-back sets. Future formulations could integrate such constraints directly to shape schedules more precisely.
- **Inclusion of game-time information**: Realistic travel and rest feasibility depend not only on games dates but also on tip-off times. Incorporating time-of-day data would allow for more accurate modeling of recovery windows and flight feasibility.
- **Distance-neighborhood experiments**: Additional experiments could explore how restricting travel to different "distance neighborhoods" influences both the objective value and the average travel burden. This analysis could clarify the trade-offs between competitive desirability and logistical efficiency.
- **Model extensions for competitive balance and star appearances**: Additional features—such as soft constraints on star-player appearances, variations in rest-period rules, or rotating fairness constraints—could be also be introduced to further enrich the scheduling framework.

These extensions would enable the scheduling model to approximate league decision-making more closely while providing increasingly refined and practically actionable schedule proposals, overall leading to an optimized schedule that benefits league GMs and athletes alike.

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